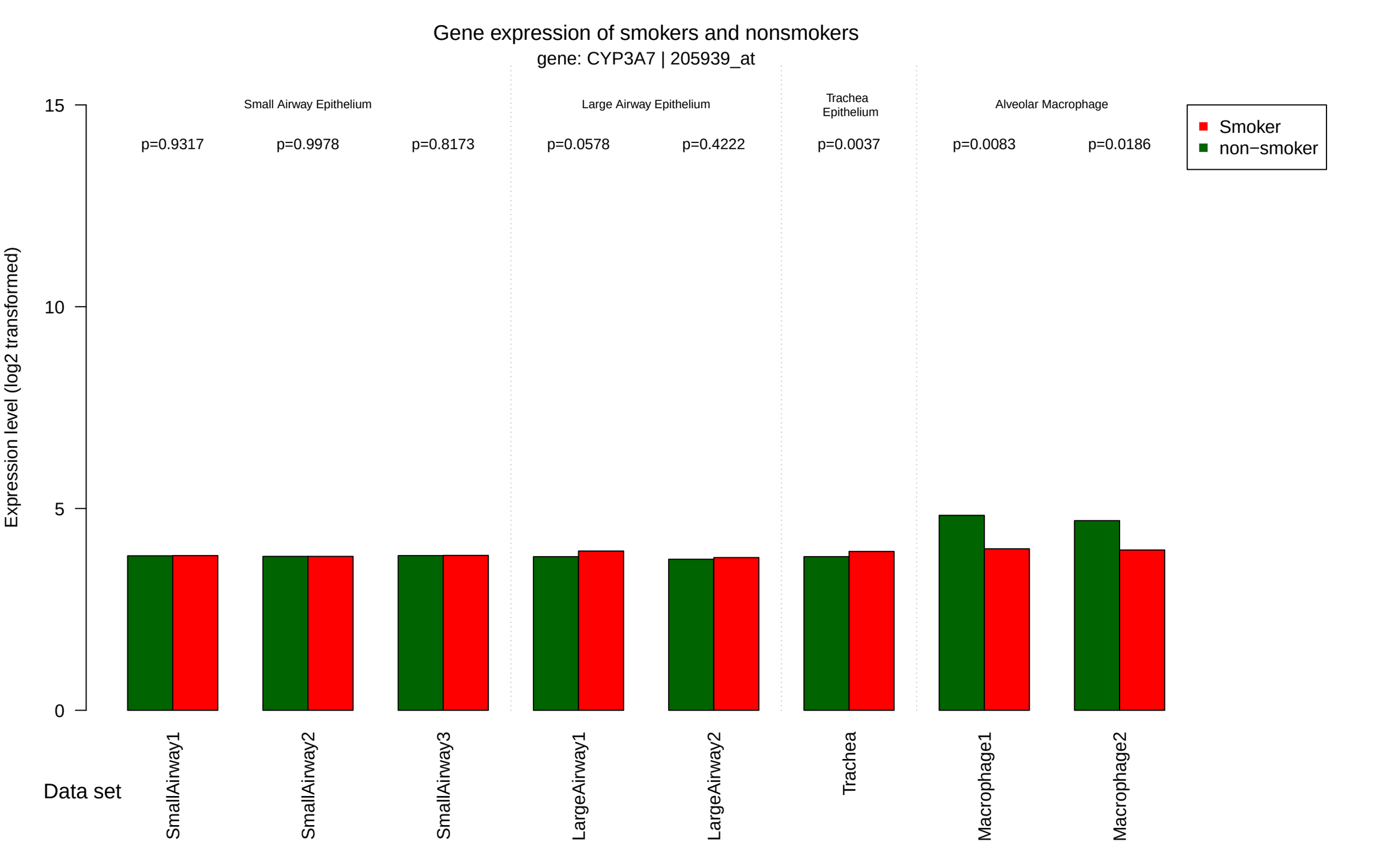
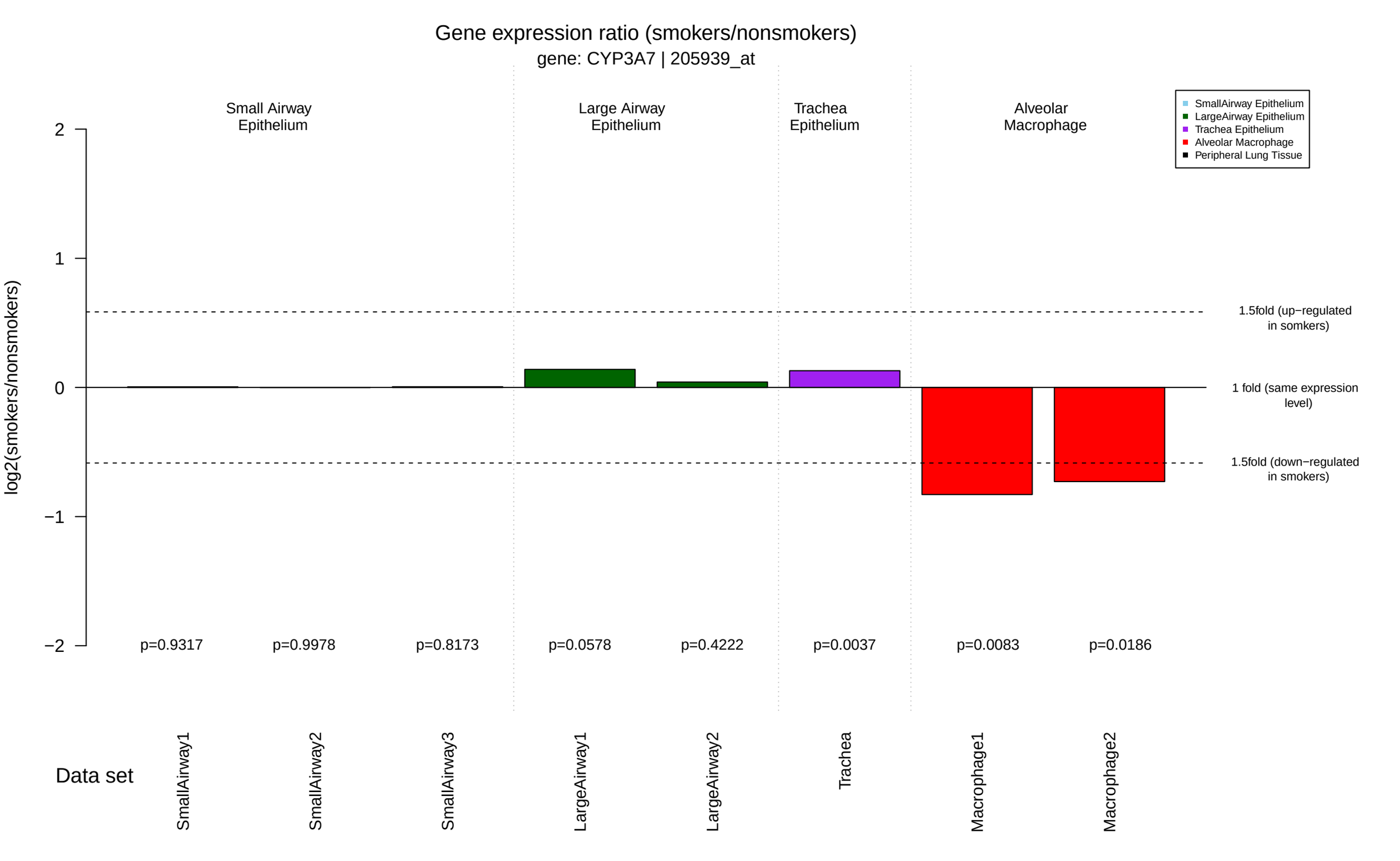
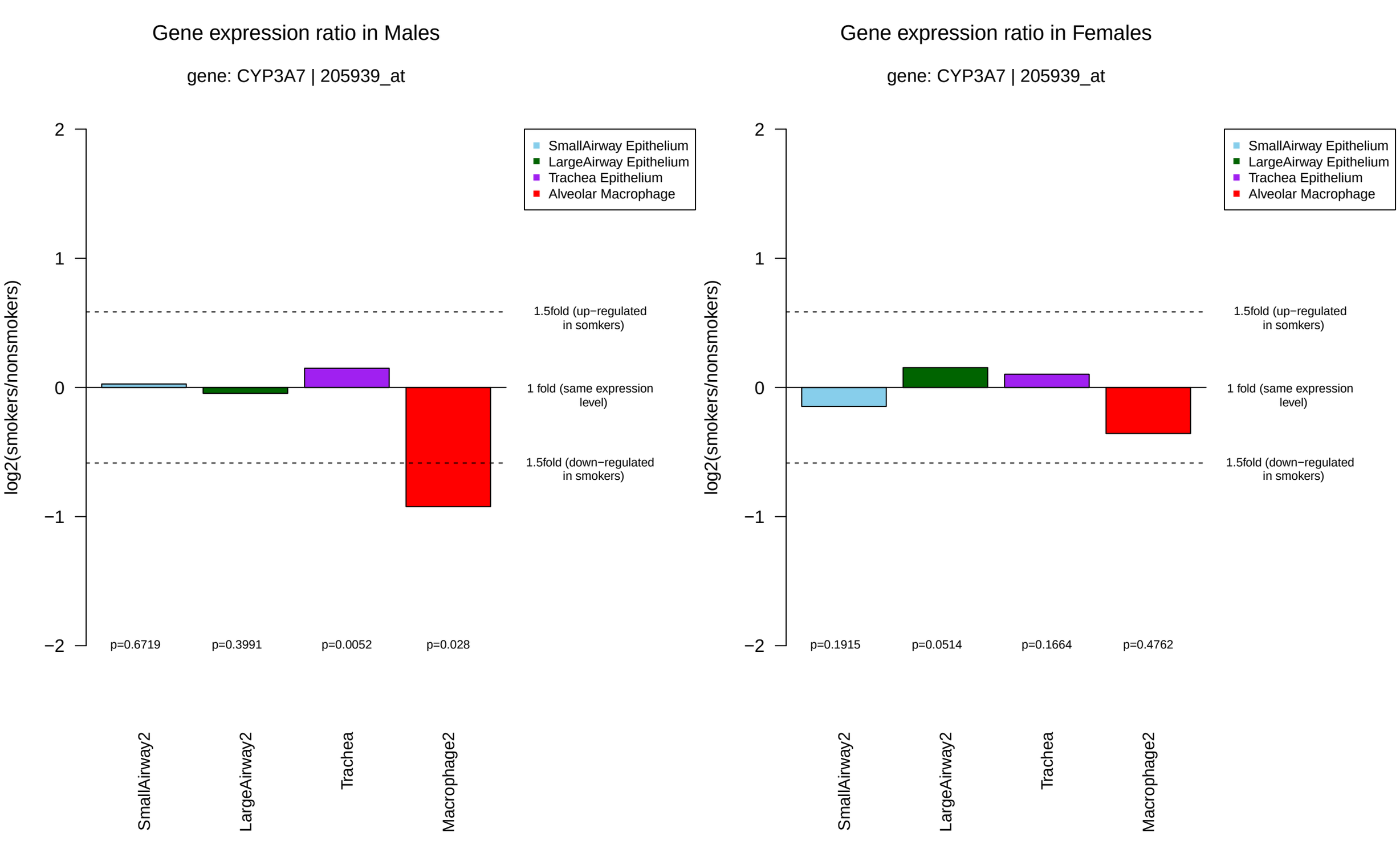
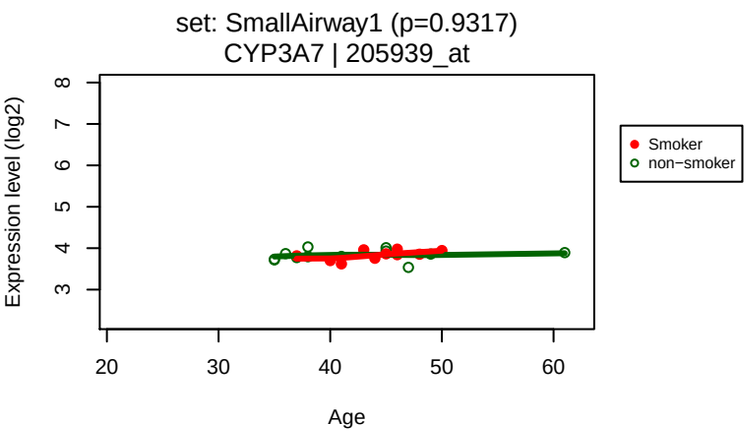
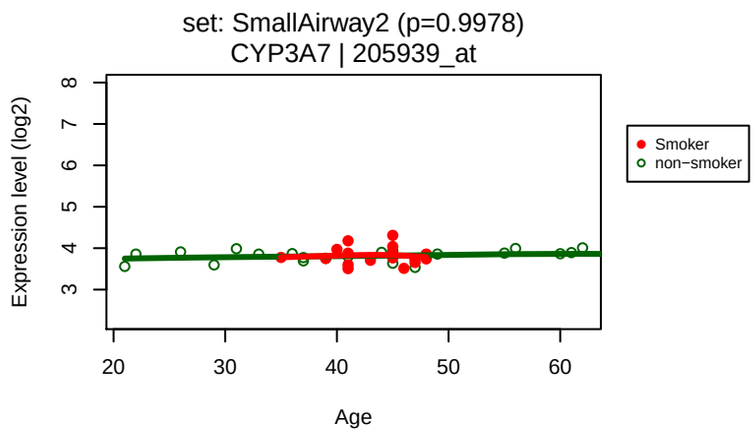
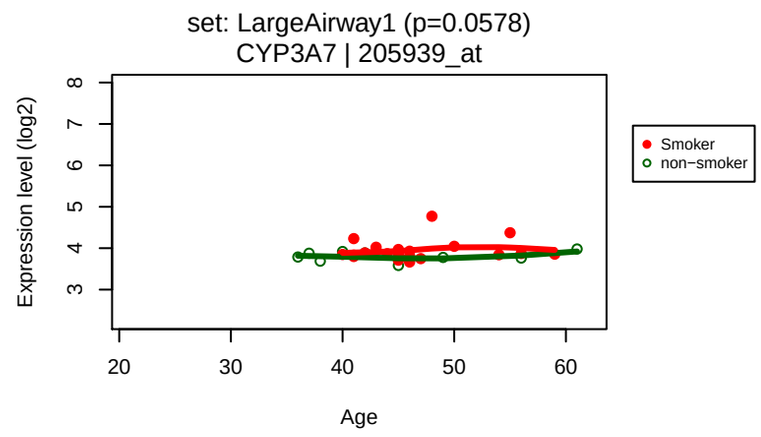

Gene expression of smokers and nonsmokers
gene: CYP3A7 | 205939_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9317
p=0.9978
p=0.8173
p=0.0578
p=0.4222
p=0.0037
p=0.0083
p=0.0186
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

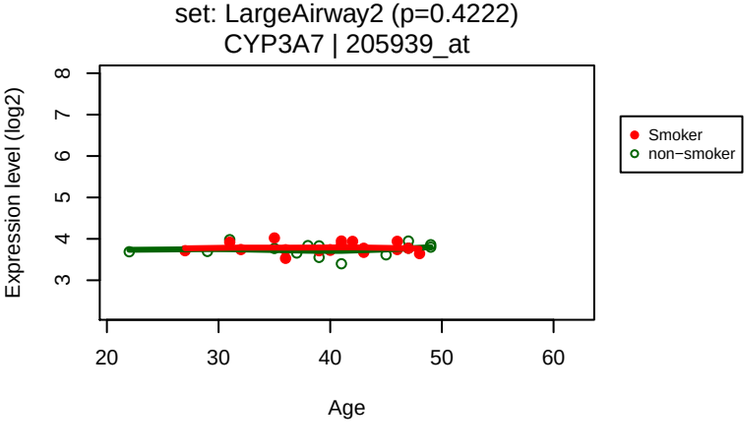

Gene expression ratio (smokers/nonsmokers)
gene: CYP3A7 | 205939_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9317
p=0.9978
p=0.8173
p=0.0578
p=0.4222
p=0.0037
p=0.0083
p=0.0186
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

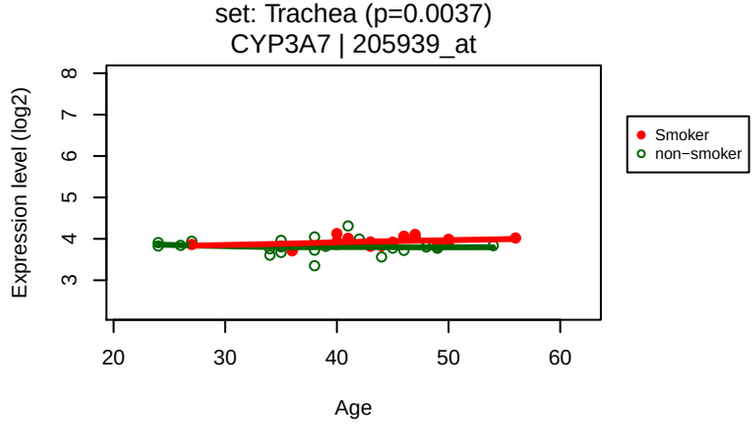

Gene expression ratio in Males
gene: CYP3A7 | 205939_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6719
p=0.3991
p=0.0052
p=0.028
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP3A7 | 205939_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1915
p=0.0514
p=0.1664
p=0.4762
SmallAirway2
LargeAirway2
Trachea
Macrophage2

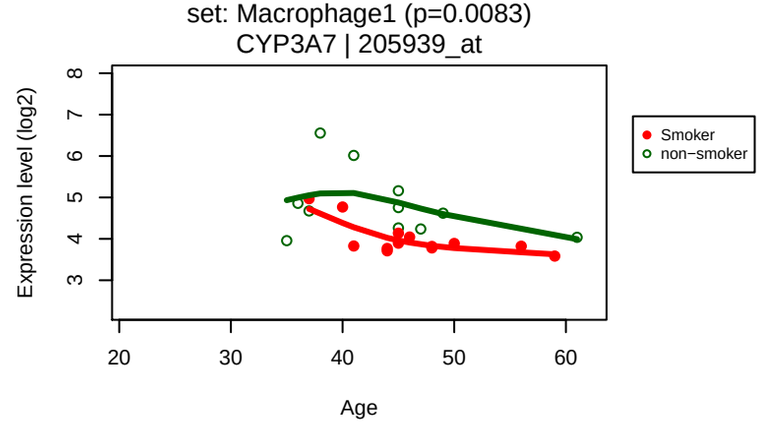

set: SmallAirway1 (p=0.9317)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker

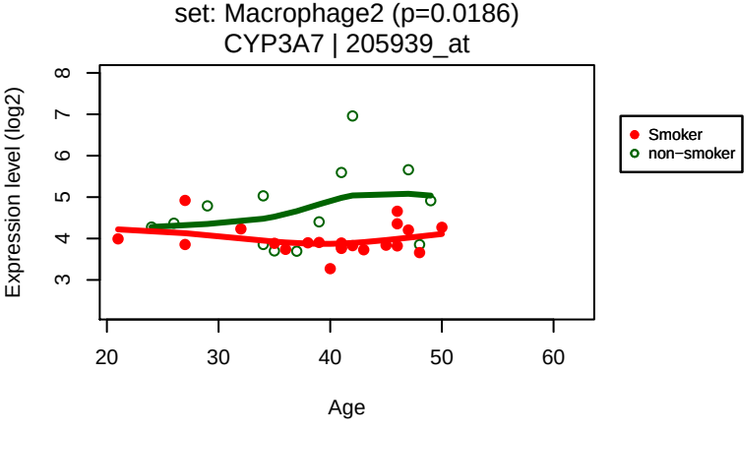

set: SmallAirway2 (p=0.9978)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker

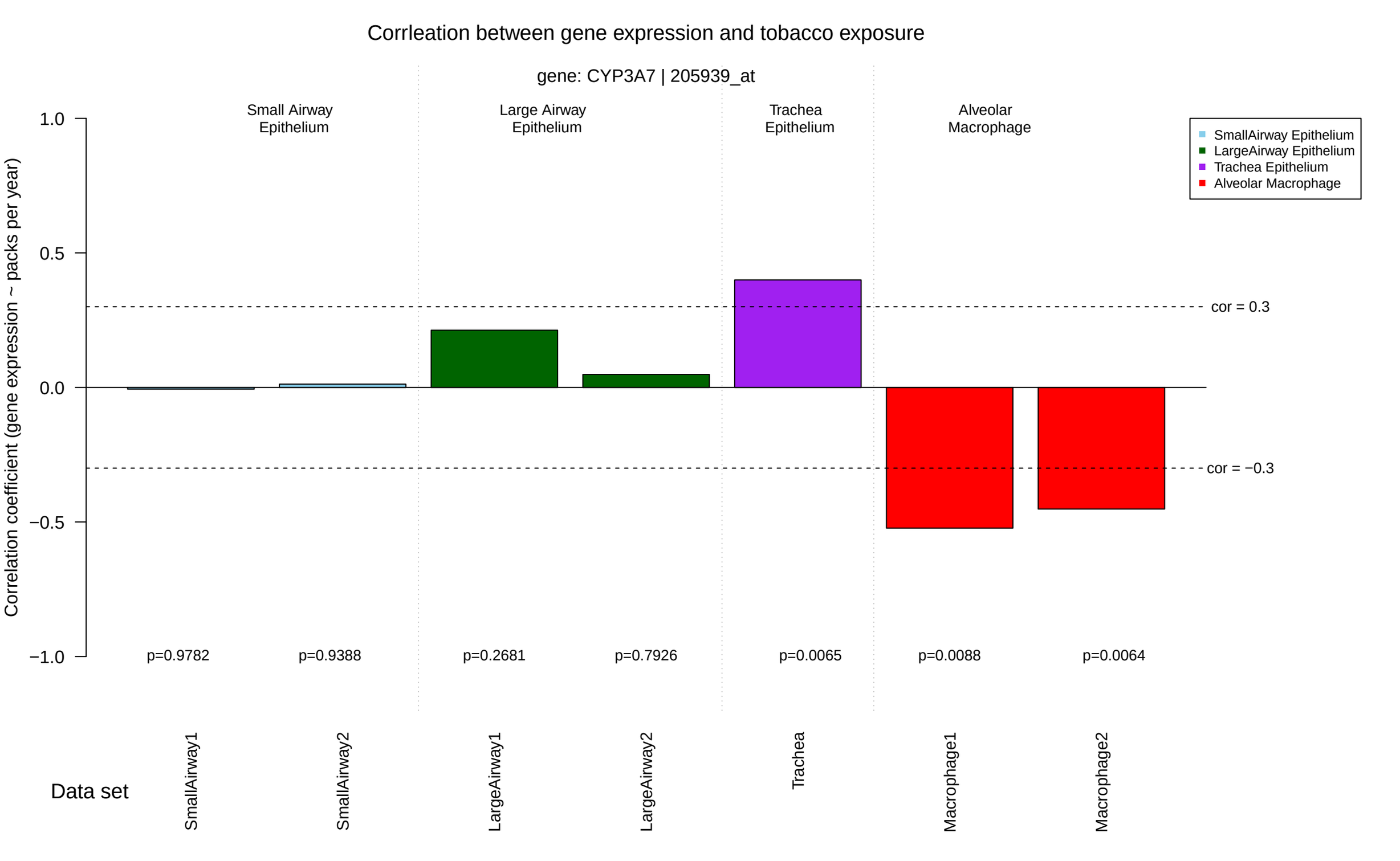

set: LargeAirway1 (p=0.0578)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4222)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0037)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0083)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0186)
CYP3A7 | 205939_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP3A7 | 205939_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9782
p=0.9388
p=0.2681
p=0.7926
p=0.0065
p=0.0088
p=0.0064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2